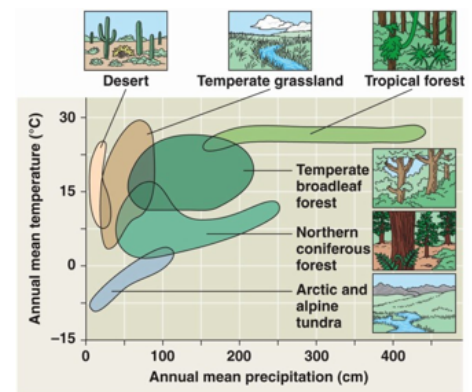


Take a deep breath. Relax. Its just 50 minutes of your life. Choose the *best* answer. Clearly *circle* the letter for your answer. Vague or unclear marks will be counted as incorrect. Each question is worth 2 points.

- The difference between communities and ecosystems is that
 - communities do *not* include interactions between individuals and the physical environment; ecosystems *do* include such interactions.
 - communities are groups of individuals of the different species; ecosystems are groups of communities.
 - communities include individuals of the different species; ecosystems are individuals of the same species.
 - communities include individuals of the same species; ecosystems include individuals of different species.
- The southeastern region of Missouri lies within the temperate broadleaf forest ecosystem. According to the climograph below, the temperate broadleaf forest

- is never warmer than the temperate grassland.
- never has more than about 220 cm (87 inches) of rainfall per year.
- always has more precipitation than the temperate grassland.
- has an average of 60–220 cm of precipitation per year.
- has more rain and similar temperatures to the northern coniferous forest.



- After transient orca began to prey on sea otters, what happened to the kelp forest?
 - The kelp abundance increased. Otters dislodge kelp with their playful antics. Because there were fewer otters, less kelp was dislodged so the kelp abundance increased.
 - The kelp abundance remained about the same. Only the predator-prey interactions between Orca and otters were important, so nothing happened to the kelp.
 - The kelp abundance decreased. Sea otters and kelp have a symbiotic relationship. Without the sea otters, the symbiotic relationship was broken and so the kelp abundance decreased.
 - The kelp abundance remained about the same. The kelp are not part of this community, so the kelp could not be affected.
 - The kelp abundance decreased. Sea otters eat urchins and urchins eat kelp. Because there were fewer otters to eat urchins, the number of urchins increased. The urchins therefore ate more kelp so the kelp abundance decreased.

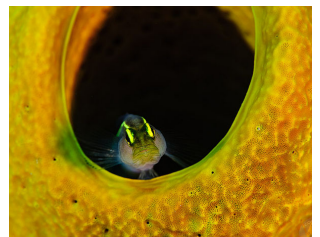
4. Higher elevations on a mountain experience less precipitation and cooler temperatures than low elevations. As you descended from the top of the mountain, you would notice a change in the types of plants and animals present in the community. This is analogous to the changes in the ecosystems if you walked
 - A. from the north pole to the south pole.
 - B. east to west over different altitudes.
 - C. west to east at one latitude.
 - D. from the north pole to the equator.
 - E. from the equator to the south pole.
 5. If emigration is greater than immigration and if the death rate is greater than the birth rate then
 - A. the population size will get larger if the birth rate is larger than immigration.
 - B. the population size will increase by exponential growth.
 - C. the population size will decrease.
 - D. the population has plenty of resources.
 - E. the population size will remain about the same.
 6. Plants produce water vapor during photosynthesis, which goes from the plant to the atmosphere. The water vapor in the atmosphere eventually falls back to the earth as rain. This process would be part of
 - A. plant metabolism.
 - B. the water cycle.
 - C. energy flow.
 - D. the energy cycle.
 - E. the entire ecosystem.
 7. If humans stopped fishing along the coast of Alaska, what will probably happen to the kelp forest and why?
 - A. The kelp abundance will remain about the same. The kelp are not part of this community, so the kelp would not be affected.
 - B. The kelp abundance will decrease. Sea otters eat urchins and urchins eat kelp. Because there were fewer otters to eat urchins, the number of urchins increased. The urchins therefore ate more kelp so the kelp abundance decreased.
 - C. The kelp abundance will decrease because more fish will be present to eat the kelp.
 - D. The kelp abundance will increase because the sea urchin abundance will probably decrease.
 - E. The kelp abundance will increase because of the symbiotic relationship between kelp and fishes. Fishes are necessary for kelp to reproduce.
-

8. Carrying capacity

- A. is the maximum population size that a particular environment can support.
- B. is seldom reached by most population because the environment has plenty of resources.
- C. is determined by birth and death rates, and emigration and immigration rates.
- D. does not for most species most of the time.
- E. describes the stress a population undergoes due to limited resources.

9. Some fish, like the goby in the picture below, live inside sponges. Sponges are a type of living organism. The goby benefits from the shelter provided by the sponge. The sponge receives no apparent benefit. As described here, this is an example of

- A. predation.
- B. parasitism.
- C. mutualism.
- D. symbiosis.
- E. commensalism.



10. Assuming a population has unlimited resources, exponential growth occurs because

- A. the birth rate exceeds the death rate.
- B. the immigration rate equals the death rate and the emigration rate equals the birth rate.
- C. the emigration exceeds the birth rate.
- D. the death rate exceeds the birth rate.
- E. the emigration rate exceeds the death rate.

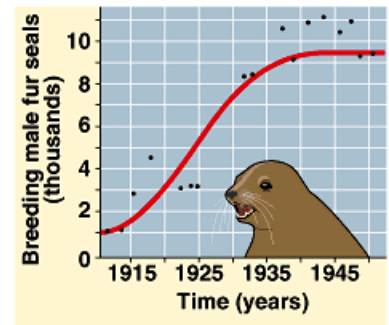
11. When the number of individuals in a population remains more or less the same over time in the specific place where these individuals live, then this population has reached the _____ of that place.

- A. maximum growth capacity
- B. carrying capacity
- C. exponential growth capacity
- D. population density
- E. logistic growth capacity

12. Considering your knowledge of the dead zone and the causes of microscopic algae blooms, the Gulf of Mexico ecosystem is most likely to be limited by which of the following nutrients?

- A. carbon
 - B. oxygen
 - C. hydrogen
 - D. CO₂
 - E. nitrogen
-

13. On average, the amount of secondary consumer biomass is about _____ less than the amount of primary producer biomass.
- A. $20\times$
 - B. $80\times$
 - C. $50\times$
 - D. $100\times$
 - E. $10\times$
14. Consider the food chain grass $\rightarrow$ grasshopper $\rightarrow$ mouse $\rightarrow$ snake $\rightarrow$ hawk. How much of the chemical energy fixed by photosynthesis of the grass (100%) is available to the hawk?
- A. 1%
 - B. 60%
 - C. 0.1%
 - D. 10%
 - E. 0.01%
15. Which of the following statements about symbiotic relationships is true?
- A. The most efficient type of parasite is one that kills its host.
 - B. Both symbiotic organisms usually benefit from the relationship.
 - C. A relationship that appears to be commensalism may in fact be mutualism or parasitism.
 - D. Neither organism benefits in a symbiotic relationship.
 - E. Both organisms are harmed in a parasitic relationship.
16. The type of growth curve shown in the figure below is a
- A. maximum growth curve.
 - B. exponential growth curve.
 - C. logistic growth curve.
 - D. population density growth curve.
 - E. carrying capacity growth curve.



17. The “role” or “job” of a species in a community, represented by all of the resources needed by the species, is called
 - A. the resource.
 - B. the niche.
 - C. a predator.
 - D. a predator or prey.
 - E. a producer or consumer.
 18. Which of the following are important biotic (living) factors that can affect the organization of communities?
 - A. predation, competition
 - B. nutrient availability, soil pH
 - C. light intensity, seasonality
 - D. precipitation, wind
 - E. temperature, water
 19. Competition occurs between two different species in a community when
 - A. one species is a predator of the other.
 - B. both species have partitioned resources.
 - C. both species are searching for mates.
 - D. one species is a parasite of the other.
 - E. both species need the same resources that is in limited supply.
 20. The difference between emigration and immigration is
 - A. the overall movement of individuals among populations.
 - B. emigration is when individuals move out of a population; immigration is when individuals move out of a community.
 - C. emigration is when individuals move into a population; immigration is when individuals move out of a population.
 - D. the direction in which individuals are moving into or out of the population.
 - E. emigration is when individuals move out of a population; immigration is when individuals move into a population.
 21. The dead zone in the Gulf of Mexico and other oceans around the globe occurs when
 - A. commercial fishers overharvest shrimp, crab, fishes and other sea animals.
 - B. an excess of nutrients causes an algal bloom. The algae consume all of the oxygen in the water for photosynthesis, so other organisms can no longer live there.
 - C. global climate change causes the water to become too warm to sustain life.
 - D. eutrophication causes an absence of nutrients in the zone.
 - E. an excess of nutrients causes an algal bloom, followed by death of the algae. Decomposition of the algae uses up the oxygen in the water, so other organisms can no longer live there.
-

22. The land on the south side of the Himalayan mountain range (below) receives relatively high levels of rainfall and has fertile farmlands. The land north of the mountain range is hot and dry and can't be farmed. Which of the following is true.

- A. Warm, moist air from the Indian Ocean blows north, rises and cools, releasing precipitation as it moves up the south side of the range, and this cool, now dry air mass heats up as it descends on the north side of the range.
- B. The desert in southern China warms the north side of the mountain range, causing it to dry out.
- C. The warm, moist air is too dense to rise over the height of Mount Everest and the other mountains in this range so the moisture remains on the south side of the range.
- D. The cool, moist Indian Ocean air heats up as it rises, releasing its precipitation as it passes the tops of the mountains, and this warm, now dry air cools as it descends on the leeward side of the range.
- E. The south side of the Himalayan mountain range is in a rain shadow.



23. Probably the most important factor(s) affecting the distribution of ecosystems is (are)

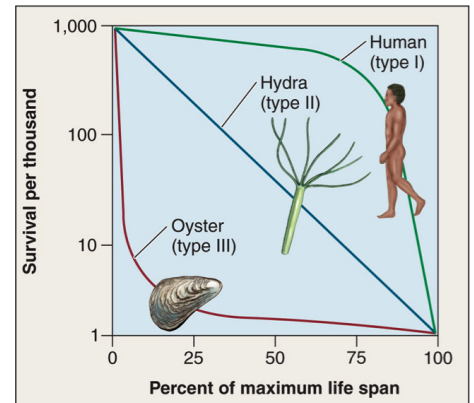
- A. longitude and latitude.
- B. climate.
- C. wind direction and rain shadows.
- D. day length and rainfall.
- E. wind and ocean water current patterns.

24. The energy that is *not* transferred to higher trophic levels is lost

- A. in fecal matter.
- B. to decomposers.
- C. to primary consumers.
- D. as metabolic heat.
- E. to secondary consumers.

25. According to the figure below, which is most correct about the offspring produced by oysters?

- A. Most offspring will die late in life.
- B. Most offspring will die at a very young age.
- C. Most offspring will die about half way through their natural life.
- D. About the same number of offspring will die every year.
- E. Most offspring will live most of their natural lifespan.



26. In 2011, the Mississippi River experienced a record flood event. Which of the following statements is the best prediction of how this record flood event will affected the Gulf of Mexico dead zone that summer?

- A. The dead zone will be smaller because the volume of water will dilute the nutrients, reducing eutrophication.
- B. The dead zone will be similar to previous years because most of the flood water will soak into the ground and not enter the Gulf of Mexico.
- C. The dead zone will be larger because the volume of water will cause more fishes to avoid the freshwater input.
- D. The dead zone will be smaller because the volume of water will bring more oxygen to the Gulf of Mexico, which will offset the hypoxia.
- E. The dead zone will be larger because the volume of water will carry even more nutrients into the Gulf of Mexico.

27. Which of the following is *not* one of the four factors that play a role in population growth rate?

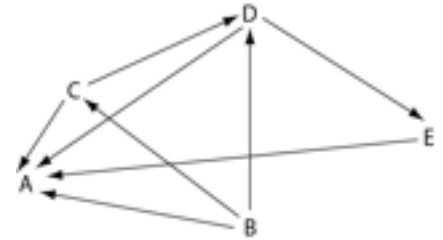
- A. Demography.
- B. Death rate.
- C. Birth rate.
- D. Immigration.
- E. Emigration.

28. Several species of darters (a group of small fishes) live together on the gravel bottoms of small streams. The darters can live in the same habitat type because they

- A. outcompeted other species of fishes, such as catfishes.
- B. created new niches within the stream.
- C. can swim at different heights above the gravel bottom.
- D. divided up the resources to reduce competition.
- E. feed at different times of the day.

29. Based on this food web, which letter represents an organism that is a primary producer? The arrows point in the direction of energy flow (to higher trophic levels).

- A. D
- B. C
- C. E
- D. A
- E. B



30. In ecosystems, why is the “cycling” used to describe the transfer of chemical elements and nutrients, but the “flow” is used for the transfer of energy?
- A. Energy flows from one ecosystem to other ecosystems, but chemical elements constantly cycle within an ecosystem.
 - B. Both chemical elements and energy flow within an ecosystem but can cycle to other ecosystems.
 - C. Both chemical elements and energy are recycled but can flow to other ecosystems.
 - D. Chemical elements are cycled into ecosystems from other ecosystems, but energy constantly flows within the ecosystem.
 - E. Chemical elements are recycled within ecosystems, but energy flows through and out of ecosystems.
-